Risk Factors and 10-Year Probability of Stroke — Framingham Heart Study

1 Introduction

The data for this analysis come from the Framingham Heart Study, a longitudinal prospective cohort study of cardiovascular disease in Framingham, Massachusetts. The dataset contains laboratory, clinic, questionnaire, and event data on 4,434 participants collected during three examination periods approximately 6 years apart (1956–1968), with up to 24 years of follow-up for stroke, myocardial infarction, angina pectoris, or death.

The goal of this analysis is to identify important risk factors of stroke and compute the 10-year probability of stroke based on different risk profiles, stratified by sex (M/F). The analysis uses only baseline (Period 1) covariate values and restricts follow-up to 10 years. As a secondary aim, we describe how risk factors change within individuals across the three examination periods to inform whether a time-varying covariate analysis might be warranted.

The scientific hypothesis is that cardiovascular risk factors measured at baseline are associated with 10-year stroke risk, and that these associations may differ between men and women. This translates to the following statistical hypotheses, tested separately for each sex: H_0: After adjusting for age, diabetes, and systolic blood pressure, candidate risk factors (prevalent CHD, BP medications, current smoking, total cholesterol, BMI) are not associated with 10-year stroke risk (hazard ratio = 1). H_A: At least one candidate risk factor is associated with 10-year stroke risk (hazard ratio $\neq$ 1).

2 Methods

2.1 Data Management

The analytic cohort was restricted to baseline (Period 1) participants free of prevalent stroke (PREVSTRK = 0), yielding 4,402 participants (1,930 men, 2,472 women). The outcome was time to first stroke within 10 years. Time to stroke was censored at 3,652 days (approximately 10 years). Deaths without stroke were tracked as a competing event; because standard Cox proportional hazards models treat these as non-informative censoring, cumulative stroke incidence may be slightly overestimated if predictors of death and stroke overlap. Missing data were handled by complete-case analysis. Antihypertensive medication use (BPMEDS) had the most missingness at approximately 1-2 percent; all other variables had less than 1 percent missing.

2.2 Statistical Analysis

Baseline characteristics were summarized by sex using means and standard deviations for continuous variables and counts and percentages for categorical variables (Table 1). Kaplan-Meier curves for stroke-free survival were estimated by sex and by candidate risk factors within sex strata to visualize unadjusted associations.

Separate Cox proportional hazards models were fitted for men and women using baseline covariate values only. A base model included three a priori established risk factors: age, diabetes status, and systolic blood pressure. Five candidate variables were then evaluated for inclusion: prevalent CHD, antihypertensive medication use, current smoking, total cholesterol, and BMI.

For model selection, AIC-based stepwise selection was used in both backward (starting from the full 8-variable model) and forward (starting from the base model) directions. The three a priori variables were force-included and could not be dropped. Running both directions provided evidence of selection robustness. Individual 1-df likelihood ratio tests at alpha = 0.10 were also reported as a secondary check. AIC-based selection was preferred because it evaluates each variable in the context of all others currently in the model, addressing the limitation that individual testing can miss variables only informative when adjusted for other covariates, and avoids the Type I error inflation of using the same data for selection and inference.

Hypotheses were tested using the Wald test on individual regression coefficients at a two-sided alpha = 0.05 level. No multiple comparison adjustment was applied. The proportional hazards assumption was assessed via Schoenfeld residuals. Model discrimination was assessed by the concordance index (C-index), ranging from 0.5 (no discrimination) to 1.0 (perfect).

To assess whether risk factors changed within individuals across the three examination periods, summary statistics were computed on a balanced cohort restricted to participants present in all three periods, so that period contrasts were not confounded by differential attrition. Within-individual paired changes (Period 3 minus Period 1) were computed for key risk factors.

3 Results

After excluding 32 participants with prevalent stroke, the analytic cohort included 4,402 participants (1,930 men, 2,472 women). Baseline characteristics are presented in Table 1. Mean age was approximately 50 years in both groups. Men had higher smoking prevalence (60.5% vs. 40.5%) and more prevalent CHD (6.2% vs. 2.7%); women had slightly higher systolic BP (133.7 vs. 131.6 mmHg). Within 10 years, 49 men (2.5%) and 62 women (2.5%) experienced a stroke, while 208 men (10.8%) and 159 women (6.4%) died without stroke, representing a substantial competing risk.

Kaplan-Meier curves stratified by sex (Figure 1) demonstrated markedly worse stroke-free survival for diabetic participants in both sexes, with the most pronounced separation among men after year 6 (Panel A). Current smokers showed modestly lower survival among men but not women (Panel B). Age 55 and older and systolic BP 140 mmHg or higher were each associated with visually lower survival in both sexes (Panels C and D).

All three variable selection methods converged on the same sets (Table 2). For men, backward elimination, forward selection, and individual LRT all retained current smoking as the only additional variable beyond the three a priori factors. For women, no candidate variable improved AIC; the final model consisted of the base variables only. This agreement across selection methods provides evidence that the model selection is robust.

The final model for men (Table 3, C = 0.797) included four variables: age (HR = 1.066 per year, 95% CI: 1.027-1.107, $p < 0.001$), diabetes (HR = 4.807, 95% CI: 2.099-11.005, $p < 0.001$), systolic BP (HR = 1.034 per mmHg, 95% CI: 1.022-1.046, $p < 0.001$), and current smoking (HR = 1.924, 95% CI: 1.036-3.573, $p = 0.038$). The diabetes hazard ratio increased from 3.535 in the base model to 4.807 after adjusting for smoking, suggesting that confounding by smoking had partially attenuated the diabetes effect.

The final model for women (Table 3, C = 0.796) included age (HR = 1.073 per year, 95% CI: 1.033-1.116, $p < 0.001$), diabetes (HR = 1.767, 95% CI: 0.636-4.904, $p = 0.275$), and systolic BP (HR = 1.026 per mmHg, 95% CI: 1.018-1.034, $p < 0.001$). Diabetes was retained as a forced a priori variable despite non-significance, reflecting the clinical rationale for its inclusion and limited statistical power from few diabetic women experiencing stroke. The forest plot (Figure 2) visually summarizes the sex-specific hazard ratios.

The Schoenfeld residual tests indicated no significant violations of the proportional hazards assumption in either model. For men, all individual terms had $p > 0.09$ (diabetes was borderline at $p = 0.097$). For women, all terms had $p > 0.70$.

Predicted 10-year stroke probabilities were computed across 15 profiles (Table 4): three ages (40, 50, 60) crossed with five risk profiles (no risk factors, high BP only, diabetes only, both, current smoker). Risk increased steeply with age and was highest for the combined high

BP plus diabetes profile. For women, the smoker profile yielded the same predictions as no-risk-factors because smoking was not retained in their model.

In the balanced cohort of 3139 participants present in all three periods, substantial within-individual changes were observed: SBP rose approximately 10 mmHg, incident diabetes developed in 6.6% of men and 5.6% of women, 21.7% of male smokers quit, and 9.9% of men and 14.7% of women started antihypertensive therapy.

4 Discussion

This analysis identified age, diabetes, and systolic blood pressure as the primary determinants of 10-year stroke risk in both sexes, with current smoking contributing additional independent risk in men only. The C-indices of approximately 0.80 indicate reasonable discriminative ability. All three selection methods agreed on the same variable sets, confirming that the individual-testing limitation of missing jointly informative variables did not materialize here. The predicted 10-year probabilities for representative risk profiles complement the C-index by providing interpretable absolute risk estimates.

Substantial within-individual changes in risk factors across the three examination periods—rising SBP, increasing diabetes prevalence, declining smoking, and growing BP medication use—indicate that baseline values become increasingly misaligned with true exposure over follow-up. A time-varying covariate Cox model might yield different results; for example, a baseline-only model may underestimate the hazard ratio for sustained smoking while misclassifying quitters as continuously exposed.

Several limitations should be noted. Standard Cox models treat deaths without stroke as non-informative censoring, potentially overestimating stroke incidence. Reported p-values should be interpreted with caution since selection and inference used the same data, though AIC-based selection partially mitigates this. Finally, the Framingham cohort was predominantly white, which may limit generalizability.

5 Reproducible Research Information

Data location: Project_3/DataRaw/frmgham2.csv
Code location: Project_3/Code/Interim_analysis.R (standalone analysis); Project_3/Code/Final_Report
(this report)
Results location: Project_3/Results/ (tables as .tex and .png; figures as .png)
GitHub repository: https://github.com/Charl3456/BIOS6624.git
Software: R version 4.5.3; key packages: survival, survminer, gtsummary, ggplot2, kableExtra

Table 1: **Table 1. Baseline Characteristics by Sex**

		Sex	
Characteristic	**Overall** N = 4,402[1]	**Men** N = 1,930[1]	**Women** N = 2,472[1]
Age (years)	49.9 (8.7)	49.7 (8.7)	50.0 (8.6)
Systolic BP (mmHg)	132.8 (22.3)	131.6 (19.4)	133.7 (24.3)
Total Cholesterol (mg/dL)	237.0 (44.6)	233.6 (42.4)	239.7 (46.2)
Missing, n (%)	52 (1.2%)	7 (0.4%)	45 (1.8%)
BMI (kg/m2)	25.8 (4.1)	26.2 (3.4)	25.6 (4.5)
Missing, n (%)	17 (0.4%)	4 (0.2%)	13 (0.5%)
Diabetes	119 (2.7%)	59 (3.1%)	60 (2.4%)
Antihypertensive Medication	136 (3.1%)	40 (2.1%)	96 (3.9%)
Missing, n (%)	60 (1.4%)	22 (1.1%)	38 (1.5%)
Current Smoker	2,169 (49.3%)	1,168 (60.5%)	1,001 (40.5%)
Prevalent CHD	187 (4.2%)	120 (6.2%)	67 (2.7%)
Stroke within 10 years	111 (2.5%)	49 (2.5%)	62 (2.5%)
Death within 10 years	413 (9.4%)	231 (12.0%)	182 (7.4%)

[1] Mean (SD); n (%)

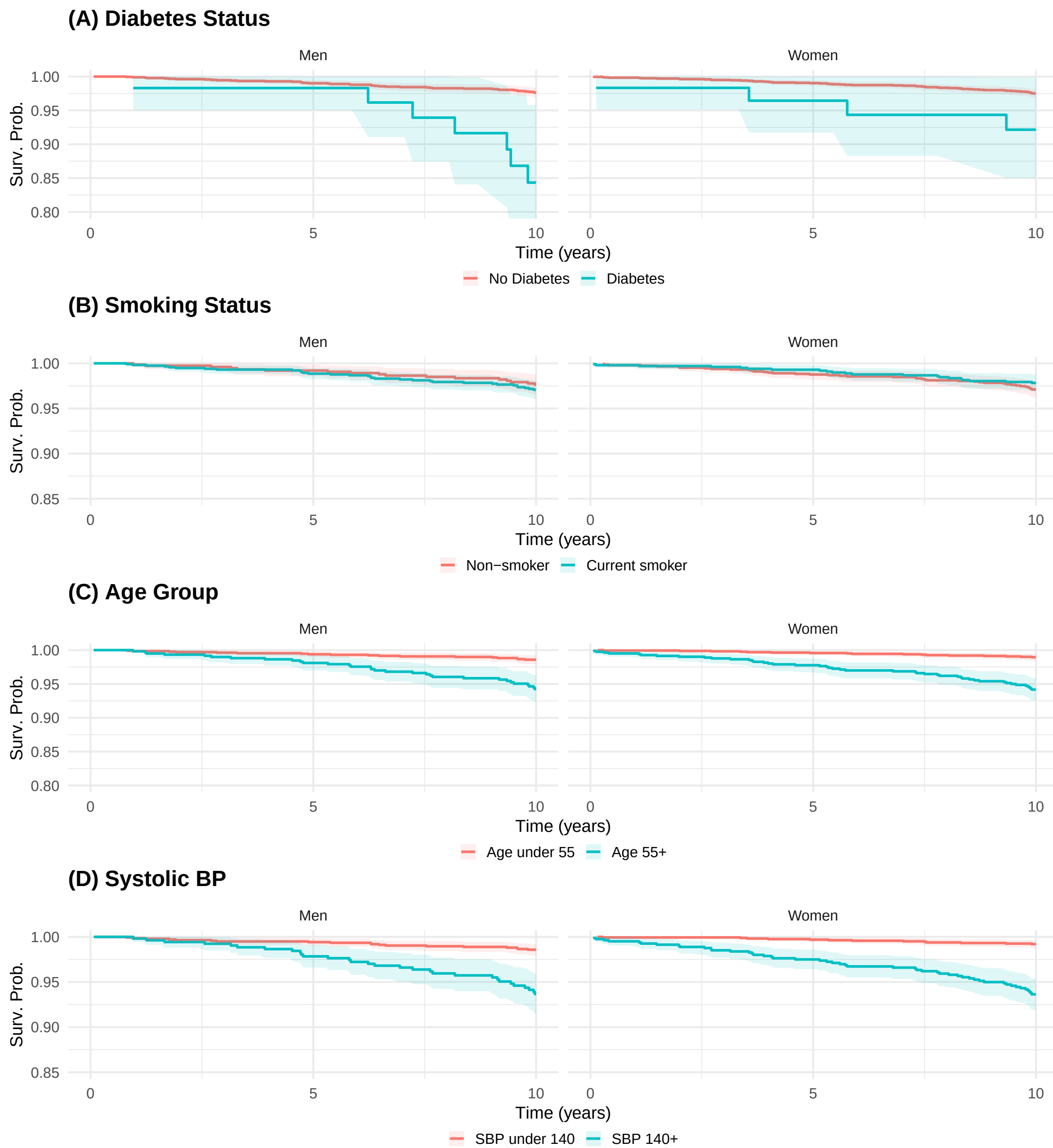

Figure 1: Figure 1. Kaplan-Meier Stroke-Free Survival Stratified by Sex: (A) Diabetes, (B) Smoking, (C) Age, (D) Systolic BP

Table 2: Table 2. Variable Selection Comparison (LRT = Individual Likelihood Ratio Test; Back = Backward AIC; Fwd = Forward AIC; M = Men; W = Women)

Variable	LRT.M	Back.M	Fwd.M	LRT.W	Back.W	Fwd.W
AGE (forced)		X	X		X	X
DIABETES (forced)		X	X		X	X
SYSBP (forced)		X	X		X	X
PREVCHD						
BPMEDS						
CURSMOKE	X	X	X			
TOTCHOL						
BMI						

Table 3: Table 3. Final Cox Models by Sex (Men C = 0.797; Women C = 0.796)

Variable	HR	95\% CI	Wald p
Men			
Age (years)	1.066	(1.027-1.107)	<0.001
Diabetes	4.807	(2.099-11.005)	<0.001
Systolic BP (mmHg)	1.034	(1.022-1.046)	<0.001
Current Smoker	1.924	(1.036-3.573)	0.0381
Women			
Age (years)	1.073	(1.033-1.116)	<0.001
Diabetes	1.767	(0.636-4.904)	0.2746
Systolic BP (mmHg)	1.026	(1.018-1.034)	<0.001

Table 4: Table 4. Predicted 10-Year Stroke Probability by Age and Risk Profile

Age	Profile	Men	Women
Age 40			
40	No risk factors	0.4%	0.5%
40	High BP (160)	1.4%	1.3%
40	Diabetes	1.8%	0.9%
40	High BP + Diabetes	6.7%	2.4%
40	Current Smoker	0.7%	0.5%
Age 50			
50	No risk factors	0.7%	1.0%
50	High BP (160)	2.7%	2.7%
50	Diabetes	3.4%	1.7%
50	High BP + Diabetes	12.4%	4.7%
50	Current Smoker	1.4%	1.0%
Age 60			
60	No risk factors	1.4%	2.0%
60	High BP (160)	5.1%	5.4%
60	Diabetes	6.4%	3.5%
60	High BP + Diabetes	22.3%	9.4%
60	Current Smoker	2.6%	2.0%

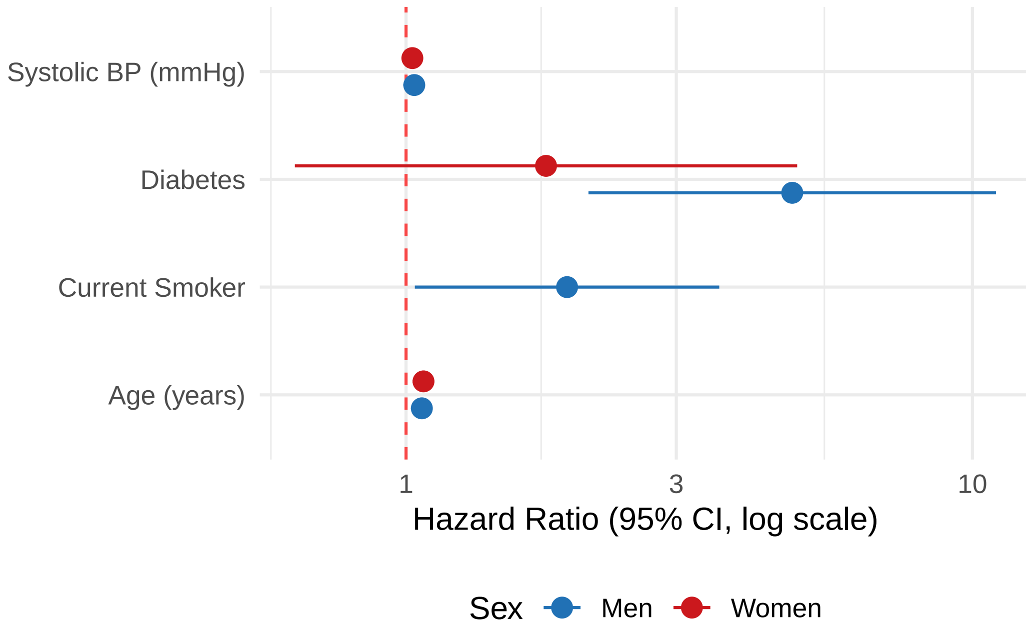

Figure 2: Figure 2. Forest Plot: Final Cox Models by Sex